Worksheet 4.2 — Concavity & the Second Derivative

Grade 12 Calculus and Vectors (MCV4U) · Unit 4: Curve Sketching

The Big Ideas

$f'' > 0$ concave up, $f'' < 0$ concave down; inflection where f'' changes sign. 2nd-derivative test: $f''(c) > 0$ min, $f''(c) < 0$ max.

- Concave up: $f'' > 0$; down: $f'' < 0$.
- Inflection: f'' changes sign.
- 2nd-derivative test classifies critical points.

The Big Picture

What concavity means

Concave up curves like a cup ($f'' > 0$); **concave down** curves like a cap ($f'' < 0$). The second derivative reads the bend.

Inflection points

Where concavity changes sign is an **inflection point** the curve switches from cupping up to down or vice versa.

The second derivative test

At a critical number, $f'' > 0$ signals a local minimum and $f'' < 0$ a local maximum.

Worked Examples

Example 1 — Always concave up

Concavity of $f(x) = x^2$.

Solution. **Step 1 (find f'):** $f'(x) = 2x$.

Step 2 (find f''): $f''(x) = 2$.

Step 3 (check the sign): $f'' = 2 > 0$ for every x — constant and positive.

Conclusion: concave up everywhere.

Example 2 — Switches

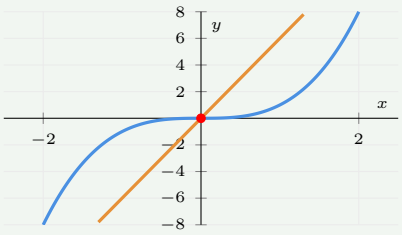
Concavity of $f(x) = x^3$.

Solution. **Step 1 (find f' and f''):** $f'(x) = 3x^2$, $f''(x) = 6x$.

Step 2 (analyze the sign of f''): for $x < 0$, $f'' < 0$ (concave down); for $x > 0$, $f'' > 0$ (concave up).

Step 3 (find where the sign changes): at $x = 0$.

Conclusion: concave down for $x < 0$, concave up for $x > 0$, inflection at $(0, 0)$:



Example 3 — 2nd-derivative test

Classify the critical point of $f(x) = x^2 - 4x$.

Solution. **Step 1 (find the critical number):** $f'(x) = 2x - 4 = 0 \Rightarrow x = 2$.

Step 2 (find f''): $f''(x) = 2$.

Step 3 (apply the 2nd-derivative test): $f''(2) = 2 > 0$, so the graph is concave up there.

Conclusion: local min at $x = 2$.

Example 4 — Inflection

Inflection point of $f(x) = x^3 - 3x^2$.

Solution. **Step 1 (find f' and f''):** $f'(x) = 3x^2 - 6x$, $f''(x) = 6x - 6$.

Step 2 (set $f'' = 0$): $6x - 6 = 0 \Rightarrow x = 1$.

Step 3 (find the y-coordinate): $f(1) = 1 - 3 = -2$.

Conclusion: inflection point at $(1, -2)$.

Example 5 — Classify all

Use f'' on $f(x) = x^4 - 2x^2$.

Solution. **Step 1 (find f' and its zeros):** $f'(x) = 4x^3 - 4x = 4x(x-1)(x+1) = 0 \Rightarrow x = 0, \pm 1$.

Step 2 (find f''): $f''(x) = 12x^2 - 4$.

Step 3 (evaluate f'' at each critical number): $f''(0) = -4 < 0$ (max); $f''(\pm 1) = 12 - 4 = 8 > 0$ (min).

Conclusion: local max at $x = 0$; local mins at $x = 1$ and $x = -1$.

Example 6 — Concave down

Concavity of $f(x) = -x^2$.

Solution. **Step 1 (find f' and f''):** $f'(x) = -2x$, $f''(x) = -2$.

Step 2 (check the sign): $f'' = -2 < 0$ for every x.

Conclusion: concave down everywhere.

Example 7 — Concave up where

Where is $f(x) = x^3$ concave up?

Solution. **Step 1 (find f''):** $f''(x) = 6x$.

Step 2 (set up the concave-up condition, $f'' > 0$): $6x > 0$.

Step 3 (solve): $x > 0$.

Conclusion: concave up on $(0, \infty)$.

Example 8 — 2nd-derivative test

Classify the critical point of $f(x) = x^2 + 6x$.

Solution. **Step 1 (find the critical number):** $f'(x) = 2x + 6 = 0 \Rightarrow x = -3$.

Step 2 (find f'' and apply the test): $f''(x) = 2 > 0$, so concave up.

Conclusion: local min at $x = -3$.

Example 9 — Inflection

Inflection point of $f(x) = x^3 - 6x^2$.

Solution. **Step 1 (find f''):** $f'(x) = 3x^2 - 12x$, so $f''(x) = 6x - 12$.

Step 2 (set $f'' = 0$): $6x - 12 = 0 \Rightarrow x = 2$.

Step 3 (find the y-coordinate): $f(2) = 8 - 24 = -16$.

Conclusion: inflection point at $(2, -16)$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Concavity of $f(x) = -x^2$?

Question 2

Where is $f(x) = x^3$ concave up?

Question 3

Use the 2nd-derivative test on $f(x) = x^2 + 6x$.

Question 4

Inflection point of $f(x) = x^3 - 6x^2$?

Question 5

At a critical point $f''(c) < 0$. What is it?

Question 6

Concavity of $f(x) = x^2 + 1$?

Question 7

Inflection point of $f(x) = x^3$?

Question 8

Use f'' to classify $x = 0$ for $f(x) = x^4 - 2x^2$.

Question 9

Where is $f(x) = x^3 - 3x^2$ concave down?

Question 10

$f''(2) > 0$ and $f'(2) = 0$: classify $x = 2$.

Question 11

Concavity of $f(x) = x^3$ at $x = -1$?

Question 12

Inflection point of $f(x) = x^3 - 3x$?

Question 13 — Challenge

Find the intervals of concavity and the inflection point of $f(x) = x^3 - 3x^2 + 1$.

Question Recap & Answer Key

Question Recap

1. Concavity of $f(x) = -x^2$?
2. Where is $f(x) = x^3$ concave up?
3. Use the 2nd-derivative test on $f(x) = x^2 + 6x$.
4. Inflection point of $f(x) = x^3 - 6x^2$?
5. At a critical point $f''(c) < 0$. What is it?
6. Concavity of $f(x) = x^2 + 1$?
7. Inflection point of $f(x) = x^3$?
8. Use f'' to classify $x = 0$ for $f(x) = x^4 - 2x^2$.
9. Where is $f(x) = x^3 - 3x^2$ concave down?
10. $f''(2) > 0$ and $f'(2) = 0$: classify $x = 2$.
11. Concavity of $f(x) = x^3$ at $x = -1$?
12. Inflection point of $f(x) = x^3 - 3x$?
13. Find the intervals of concavity and the inflection point of $f(x) = x^3 - 3x^2 + 1$.

✓ Answer Key

1. concave down everywhere
2. $x > 0$
3. min at $x = -3$
4. $(2, -16)$
5. local max
6. concave up
7. $(0, 0)$
8. max ($f'' = -4 < 0$)
9. $x < 1$
10. local min
11. concave down
12. $(0, 0)$
13. $f'' = 6x - 6$: down on $x < 1$, up on $x > 1$, inflection $(1, -1)$